Software Engineering Lab/ 3th Grade

[Second Semester] and [2026]

[Lab 4] [24/2/ 2026] [8:30 + 10:30]

Instructor Information

Instructor	Email	Hours
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LOOPING

Looping refers to the repeated execution of statements. This technique comes in very handy because it means that you can repeat operations as many times as you want (thousands, even millions, of times) without having to write the same code each time.

do Loops

do loops operate as follows. The code you have marked out for looping is executed, a Boolean test is performed, and the code executes again if this test evaluates to true, and so on. When the test evaluates to false, the loop exits.

The structure of a do loop is as follows, where <Test> evaluates to a Boolean value:

```
do
  {
        <code to be looped>
  } while (<Test>);
```

For example, you could use the following to write the numbers from 1 to 10 in a list:

```
int i = 1;
do
  {
        MyList.add (i);
        i++;
  } while (i <= 10);
```

Exercise:

Read numbers from tow Textbox then add these numbers to the number list as shown (Use your Previous expertise)

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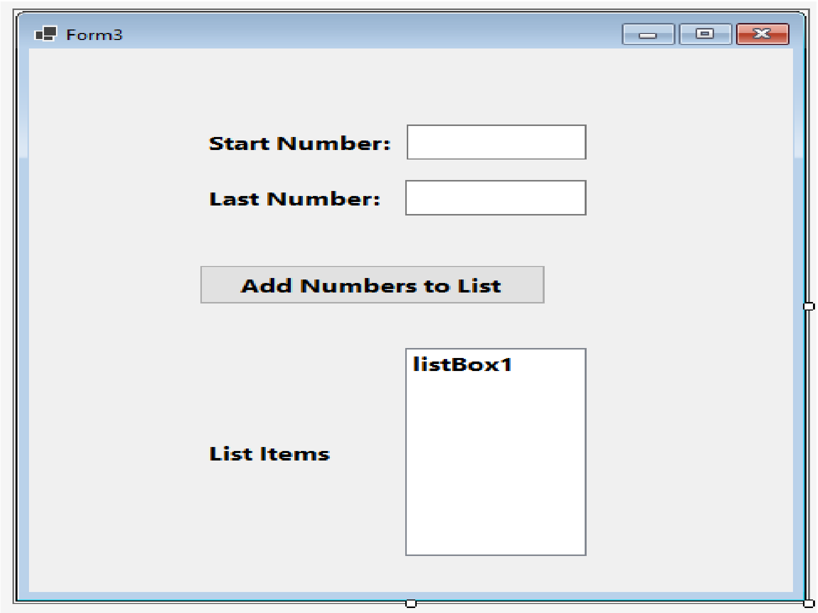

while Loops

while loops are very similar to do loops, but they have one important difference: The Boolean test in a while loop takes place at the start of the loop cycle, not at the end. If the test evaluates to false, then the loop cycle is never executed. Instead, program execution jumps straight to the code following the loop.

Here's how while loops are specified:

```
while (<Test>)
  {
     <code to be looped>
  }
```

They can be used in almost the same way as do loops:

```
int i = 1;
while (i <= 10)
{
MyList.add (i);
}
```

Exercise:

Read numbers from two Combobox then add the odd numbers to the odd list and add the even numbers

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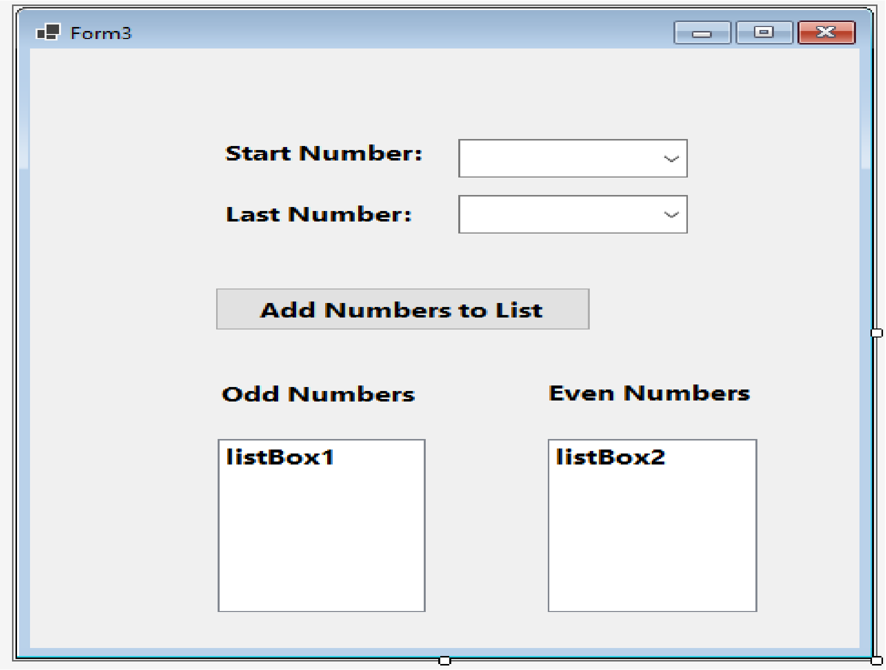
to the even list as shown (Use your Previous expertise)



for Loops

The last type of loop to look at in this chapter is the for loop. This type of loop executes a set number of times and maintains its own counter. To define a for loop you need the following information:

- ➤ A starting value to initialize the counter variable
- ➤ A condition for continuing the loop, involving the counter variable
- ➤ An operation to perform on the counter variable at the end of each loop cycle

For example, if you want a loop with a counter that increments from 1 to 10 in steps of one, then the starting value is 1; the condition is that the counter is less than or equal to 10; and the operation to perform at the end of each cycle is to add 1 to the counter.
This information must be placed into the structure of a for loop as follows:

```
for ( <initialization> ; <condition> ; <operation> )
{
   <code to be looped.
}
```

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This works exactly the same way as the following while loop:

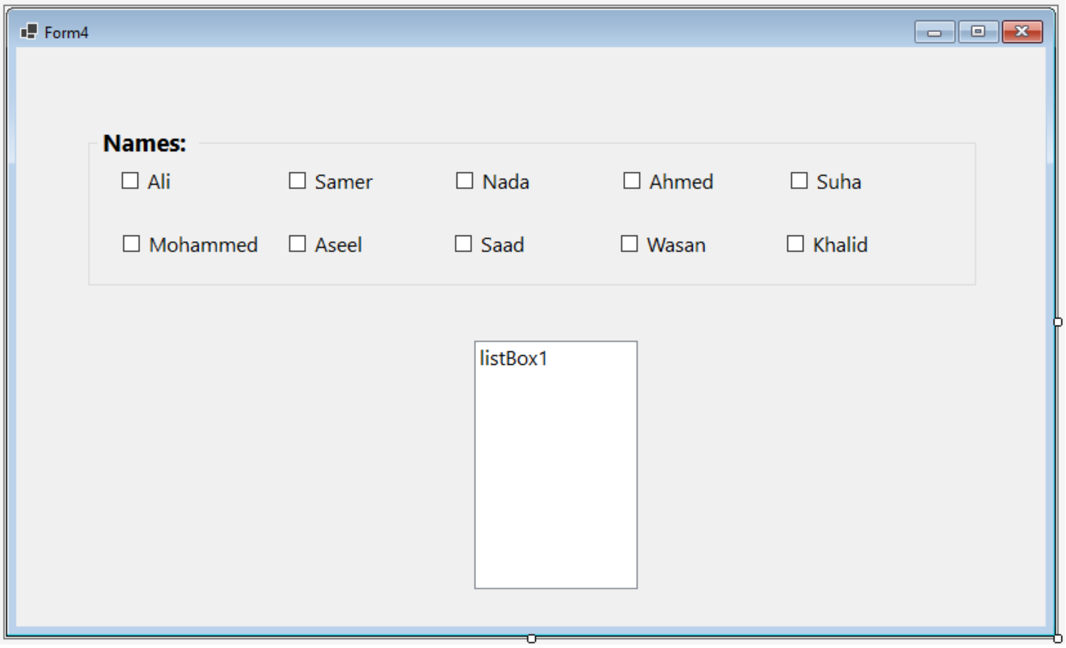
```
<initialization>
while (<condition>)
{
    <code to be looped>
    <operation>
}
```

Earlier, you used do loops and while loops to write out the numbers from 1 to 10. The code that follows shows what is required to do this using a for loop:

```
int i;
for (i = 1; i <= 10; ++i)
{
   MyList.add(i);
}
```

Homework 1:

Use your Previous expertise in Programming to add or remove these names to or from list.



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Homework 2:

Use your Previous expertise in Programming to select several names then add or remove them to or from list1; after that; moves these names between two lists.